

Architecture Selection

We chose Layered Architecture for the EV Charging Station Reservation System.

Architectural Choice: Layered Architecture, with a Presentation Layer (Driver App UI, Operator Console UI), a Business Layer (Reservation Manager, Station & Socket Manager, Pricing & Penalty Service, Notification Service) and a Data Layer (Database Component).

Two Reasons

- Clear separation of concerns: The EV Driver and Station Operator need different screens, but both depend on the same rules, such as connector compatibility, slot availability and dynamic pricing. In a layered design these rules sit once in the Business Layer, and both UIs call them through interfaces such as IReservation and IStationAdmin.
- Consistent reservation data: FR-001 requires that two vehicles are never booked on the same socket at the same time, and NFR-001 requires releasing a socket after a no-show. Keeping all reservation and socket data behind one Data Layer makes it easier to keep this data consistent, unlike a microservices design with separate data stores.

Security Advantage: The UIs can reach the database only through the Business Layer, so every request passes through the Reservation Manager and its checks first. Authentication and validation are applied in one place, and a driver or operator never gets direct access to stored reservation, pricing or station data.

Performance Benefit: Availability queries (NFR-002, displayed within 2 seconds) run inside a single deployed application, so a request moves between layers as in-process calls with no network hops between services. This keeps response time low compared with a microservices design, where one request may cross several network calls.